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Project Description PP2 WS2026/27

EAT: A Benchmark for Annotation Transfer with Raw pLM Embeddings

1 Background

To annotate an uncharacterised protein, we usually copy the annotation of a similar, already characterised one. Embedding-based annotation transfer (EAT) does this in the embedding space of a protein language model (pLM): a frozen pLM turns every protein into one vector, and a query inherits the annotation of its nearest neighbour among the labelled proteins (the *lookup set*). The original tool, goPredSim, turns the Euclidean distance d to that neighbour into a reliability index, $\mathrm{RI} = 0.5/(0.5 + d)$ (1 for identical embeddings). Without any training, EAT transferred Gene Ontology terms better than BLAST[1], and for remote homologues, raw ProtT5 embeddings nearly doubled the accuracy of the sequence search MMseqs2 at the CATH superfamily level (64 vs. 35%; profile HMMs reached 77%)[2].

Since then, many new pLMs have appeared (e.g., ProstT5, ESM-2, ESM C, Ankh), and the choice matters: on a strict CATH benchmark, swapping frozen ProtT5 for frozen ProstT5 embeddings raised superfamily accuracy from 69.6 to 78.5%[3]. Papers on new pLMs, or on methods trained on top of embeddings, often report raw EAT as a baseline, but each sets it up differently (pLM, pooling, distance, number of neighbours, lookup set) and rarely documents it fully: raw ProtT5 reaches 64% in one paper and 73% in another at the superfamily level, on the same CATH test and lookup sets[2,4]. Finally, the RI cutoffs that decide whether a transfer can be trusted were set for an older pLM (SeqVec), and RI values are not comparable between pLMs, whose distances live on different scales[1].

For sequence alignments, HFSP[5] answers when transfer is safe: a curve over sequence identity and alignment length above which two enzymes likely share their EC number (precision 96% for the first three EC levels, but 44% for all four). In short, raw EAT has a de facto CATH test set, but no shared protocol across pLMs and tasks, and no distance thresholds calibrated on held-out data, i.e., no HFSP for embeddings. In this project, you will build this missing yardstick: a transparent benchmark of raw embeddings from current pLMs for protein structure (CATH,

[1] Littmann, M., Heinzinger, M., Dallago, C., Olenyi, T. & Rost, B. “Embeddings from Deep Learning Transfer GO Annotations beyond Homology.” *Scientific Reports* 11, 1160 (2021). https://doi.org/10.1038/s41598-020-80786-0.

[2] Heinzinger, M., Littmann, M., Sillitoe, I., Bordin, N., Orengo, C. & Rost, B. “Contrastive Learning on Protein Embeddings Enlightens Midnight Zone.” *NAR Genomics and Bioinformatics* 4(2), lqac043 (2022). https://doi.org/10.1093/nargab/lqac043.

[3] Miller, D., Bordin, N., Jeyananthan, J., Waman, V., Heinzinger, M. & Orengo, C. “ContrasTED: Contrastive Domain Embeddings for Scalable Remote Homology Classification.” *bioRxiv* (2026). https://doi.org/10.64898/2026.08.28.747864.

[4] Elnaggar, A., Essam, H., Salah-Eldin, W., et al. “Ankh: Optimized Protein Language Model Unlocks General-Purpose Modelling.” *arXiv* 2301.06568 (2023). https://doi.org/10.48550/arXiv.2301.06568.

[5] Mahlich, Y., Steinegger, M., Rost, B. & Bromberg, Y. “HFSP: High Speed Homology-Driven Function Annotation of Proteins.” *Bioinformatics* 34(13), i304–i312 (2018). https://doi.org/10.1093/bioinformatics/bty262.

which classifies domains by their 3D structure) and enzyme function (EC numbers, which classify enzymes by the reaction they catalyse), which anyone can use to check whether, and by how much, a new pLM, or a method trained on top of one, beats raw EAT. As nothing is trained in raw EAT, its result reflects only how a pLM arranges proteins, with no trained head to hide differences: the benchmark is also a fair way to compare pLMs.

Research questions

1. Which pLM and which distance measure (e.g., Euclidean or cosine) transfer structure and function annotations best, and where does EAT beat plain sequence similarity?
2. Up to which distance is a transfer safe (e.g., $\geq$95% precision), for structure and for function? Can distance be turned into a reliability score that means the same for every pLM (e.g., 0.9 = 90% of such transfers are correct)?

Scope. "Raw" means: the pLM stays frozen, each protein is represented by the mean of its per-residue embeddings from the last layer (layer and pooling are fixed), nothing is trained on top of its embeddings (no contrastive or metric learning, no probes, no fine-tuning), and a query takes the label of its nearest neighbour(s). Label-free steps such as centring are allowed. Annotations may be used on a calibration set (i.e., a validation set) to choose settings, set thresholds and fit the reliability score, never on the test set. pLMs pre-trained with annotations (e.g., ESM3) form a separate group. Learned methods such as ProtTucker, ContrasTED or CLEAN are reference points: compare their numbers only with the raw baseline of the same paper.

Minimum result. For ProtT5, ProstT5, ESM-2 (8M, 35M, 150M, 650M, 3B), ESM C (300M, 600M), Ankh-base, Ankh-large and Ankh3-large:

- accuracy per level on CATH and EC, with confidence intervals, next to transfer from the best MMseqs2 hit, binned by sequence identity to the lookup set;
- distance thresholds per pLM, task and level at the precision(s) you state, checked on test data;
- an updated reliability score;
- documented code and frozen splits that score any new pLM or method the same way.

2 Datasets

- **CATH (structure)** sorts protein domains into four levels: Class, Architecture, Topology, Homologous superfamily (C/A/T/H). Start with the ProtTucker benchmark (`lookup69k`, `test219`) and EAT code (https://github.com/Rostlab/EAT); the pre-computed CATH-S100 lookup in that repository contains all `test219` domains, so never use it as lookup for them. Then use the larger non-redundant CATH v4.4 domain sets (http://download.cathdb.info/cath/releases/all-releases/v4_4_0/non-redundant-data-sets/).
- **EC numbers (function)** classify enzymes in four levels (e.g., EC 3.4.21.4; incomplete: EC 3.4.-.-). There is no ready-made split. Two starting points: (i) UniProtKB entries with experimental evidence for their catalytic activity: in release 2026_03, the query `(ec:*) AND (cc_catalytic_activity_exp:*)` at https://www.uniprot.org finds 48,401 entries, all in Swiss-Prot (which has 279,671 entries with an EC number; unreviewed entries have no



such evidence); (ii) the HFSP set[5] of 7,022 Swiss-Prot enzymes (release 2017_09) with one complete EC number, experimental evidence of protein existence and an EC annotation in BRENDA (accessions in the paper's Supplementary Data 1). Older UniProt releases, e.g., for time-based splits: https://ftp.uniprot.org/pub/databases/uniprot/previous_releases/.

- **Embeddings**: the ProtSpace embedding notebook covers all pLMs above except ESM-2 3B; for that one, load the Hugging Face checkpoint `facebook/esm2_t36_3B_UR50D` with `transformers`. The biocentral API (https://github.com/biocentral/biocentral_api) is an alternative, but it currently returns wrong ESM C embeddings, so do not use it for ESM C. Compute all embeddings of one pLM with the same tool, and time a small batch first: the EC lookup set is the compute bottleneck.

3 Exploration Steps

1. **Get started.** Read the EAT, ProtTucker and HFSP papers[1,2,5]. A natural first step is to reproduce raw ProtT5 on `test219` (84/67/57/64% at C/A/T/H). If your numbers differ, find out why.
2. **Design a fair benchmark.** Draft answers to the questions below, agree with your supervisors on splits and protocol, then freeze them.
   - How do you split lookup and query proteins so that results reflect real use? How similar are queries to the lookup set?
   - Which EC labels can you trust as test labels? (Most EC numbers in UniProtKB are inferred, not measured, and evidence is given per reaction, so one entry can mix measured and inferred EC numbers.)
   - From ESM-2 8M to 3B: is a larger model better because it knows more, or because its embeddings have more dimensions?
   - When can Euclidean and cosine distance pick different neighbours at all? How do you compare pLMs whose distances live on different scales?
   - What happens with long or multi-domain proteins, and with enzymes that have several or incomplete EC numbers?
   - Besides the best MMseqs2 hit, which baselines must EAT beat to be useful?
3. **Answer the research questions** with your frozen protocol, then go beyond the minimum with your own questions, e.g., normalising embeddings, or where and why EAT fails.

